

NISHAN PAUL

AI Engineer

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Dhaka, Bangladesh in linkedin.com/in/nishanpaul2022 github.com/nishan-paul-2022



EXPERIENCE

Dreams For Tomorrow | 2023-2023 | Contract Developer

Built a multi-tenant Laravel CMS and blogging platform, enabling secure resource sharing and collaborative forums for educational environments.

Tech Cloud Ltd. | 2025-2025 | Internship

Architected a Laravel-based accounting ecosystem, automating real-time financial tracking and reporting to maximize operational accuracy.

PROJECTS

CMatrix - Agentic Red Team

FastAPI Next.js LangGraph LangChain PostgreSQL Qdrant Celery

Agentic red team for running controlled, human-in-the-loop security simulations across modern applications and infrastructure. It combines multi-agent orchestration, approval gates, monitoring, and audit-friendly workflows to help teams test defenses safely, repeatably, and at scale. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under [cmatrix.kaiofficial.xyz](#)

OpenCV Visualizer

Python OpenCV NumPy Computer Vision Image Processing Tkinter

A comprehensive toolkit for Computer Vision mastery using OpenCV and Python. This repository features 28+ modular implementations of image processing techniques, feature detection algorithms, and real-time video analysis tools (including Lane Detection & Invisible Cloak).

Barcodey - Barcode Scanner for Anything

Next.js NestJS PostgreSQL Redis TypeORM Socket.io BullMQ

Universal Barcode Intelligence & Scanning Suite. High-performance monorepo for scanning and analyzing barcodes at scale. Features real-time sync, Tailscale private networking, and Docker containers. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under [barcodey.kaiofficial.xyz](#)

EDUCATION

BSc in CSE | 2021 – 2026
Chittagong University of Engineering and Technology (CUET)
Chittagong, Bangladesh

HSC in Science
Notre Dame College
Dhaka, Bangladesh

SSC in Science
Motijheel Govt. Boys' High School
Dhaka, Bangladesh

AI ENGINEERING

LangGraph LangChain PyTorch
Transformers Agentic AI
Deep Learning FastAPI Qdrant
Celery

AI TOOLS

Antigravity Cursor Kilo Code
Supermaven Gemini CLI Qwen CLI

DEVOPS

Docker Docker Compose
Containerization
CI/CD (GitHub Actions) Linux Systems
Nginx / Reverse Proxy
VPS Deployment SSL/TLS (Certbot)

LANGUAGES

JavaScript (ES6+) TypeScript
Python Java C# C/C++

PROJECTS

Markify - Markdown to PDF Converter

Next.js Zustand Framer Motion Shadcn UI PostgreSQL Docker

A simple and fast Markdown-to-PDF converter that transforms .md files into clean, professional PDF documents. Supports code formatting, structured content, and customizable output for developers. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under markify.kaiofficial.xyz

Flick - P2P File Sharing

Next.js React TypeScript Tailwind CSS WebRTC PeerJS

Instant peer-to-peer file sharing app built with Next.js and WebRTC. No server storage, no accounts — just share a room code and transfer files directly between devices. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under flick.kaiofficial.xyz

PROBLEM SOLVING

Dynamic Programming (10) Greedy (38) Strings (16)

Hashing (2) Bit Manipulation (2) Graphs (3)

Algorithms (27) Data Structure (4)

Math (54) Implementation (89)

- [Codeforces](#) | 236 solved
- [LightOJ](#) | 89 solved
- [Virtual Judge](#) | 81 solved
- [UVA](#) | 14 solved

ACHIEVEMENT

Vivasoft AI Hackathon 2025

🐾 Team: KAI

🏆 Ranking: Top 10

FRONTEND

Next.js React Tailwind CSS

Shadcn UI Zustand Framer Motion

BACKEND

NestJS Node.js Redis TypeORM

Socket.io BullMQ Swagger

DATABASE

PostgreSQL SQL Server SQLite

DEV TOOLS

Git / GitHub

Pre-commit Hooks (Husky)

Build Tools (Vite / Turborepo)

Secure Auth (OAuth 2.0 / JWT)

TESTING

Automated Testing (Jest) Postman

ESLint Prettier

Mesh Networking (Tailscale)

CERTIFICATES

- [Managing Machine Learning Projects with Google Cloud](#)
- [AI For Everyone](#)
- [Getting Started with Python](#)
- [Python Data Structures](#)

BLOGS

- [Backtracking Technique & N Queen Problem](#)
- [Tiling a Defective Chessboard Using Divide & Conquer Technique](#)
- [Setup Flutter On VScode](#)